

Arjun Sood

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Education

Georgia Institute of Technology | Atlanta, GA

Present

Bachelor of Science in Computer Engineering, GPA 3.81

Expected Graduation, May 2028

Activities: Robotics Research @ RICAL Lab, RoboJackets RoboCup Electrical Subteam, AI @ GT Applied Research Project Team, GT IEEE, and Startup Exchange

Skills

Robotics & AI: ROS 2, Gazebo, RViz, SLAM, EKF (Sensor Fusion), Vision-Language-Action Models, Reinforcement Learning, AI Agents, Path Planning (A*, RRT), Autonomous Navigation, Simulation-to-Real

Programming: Python, C/C++, Java, PyTorch, OpenCV, MATLAB, React, Django, FastAPI, HTML/CSS, Figma

Embedded & Hardware: Arduino, Dynamixel, SPI/I²C/UART, Motor Drivers, Power Distribution, KiCad PCB Design, Multimeters

Systems & Tools: Linux, Git, Docker, Conda, Bash, ROS2 Package Development

Experience

Georgia Institute of Technology RICAL Robotics Laboratory | Atlanta, GA

August 2025 - Present

Robotics Engineer & Researcher

- Designed a 4-wheel drive robotic module using Arduino Mega and Dynamixel motors with wireless control
- Developed a modular ROS 2 package for reusable wheel control and autonomy integration
- Integrated wheel system onto a quadruped robot, extending hybrid locomotion capabilities
- Implemented vision-based obstacle detection and autonomous path planning
- Documented hardware and software architecture for reproducible robotics research

Color Robotics | San Francisco, CA

March 2025 – October 2025

Software Engineering Intern

- Built an LLM-driven diagnostic agent using RAG and knowledge graphs to analyze robot faults, recommend actions, and generate structured maintenance reports
- Developed robot monitoring dashboards displaying live performance metrics, errors, and deployments using React, Django, and Python
- Collaborated with robotics engineers to bridge backend telemetry with operator-facing tools

University of Pennsylvania ASSET Laboratory | Philadelphia, PA

June 2024 – August 2025

AI/ML Engineer & Researcher

- Conducted research on LLMs and video foundation models under Prof. Mayur Naik
- Co-led training of an LLM using Group Relative Policy Optimization to generate autonomous Python workflows
- Developed a selective fine-tuning and quantization method achieving near-full performance with 50% fewer non-zero bits
- Proposed efficiency improvements for video-based object-action detection systems

Selected Project

Simulation of Autonomous TurtleBot in a Dynamic Maze

Fall 2025

Designed and implemented a ROS2-based TurtleBot simulation in Gazebo that autonomously navigates dynamic mazes

- Built a ROS 2 + Gazebo simulation enabling autonomous navigation in dynamic mazes
- Implemented SLAM, EKF localization, and custom path planning (A, RRT)*
- Integrated OpenCV-based perception for goal detection and recovery behaviors
- Developed reproducible pipelines using Python/C++, Docker, and Linux

Activities

Georgia Tech's RoboJackets Robocup Team | Atlanta, GA

August 2025 - Present

Electrical Engineer

- Designing robot power systems, PCBs, and embedded firmware for autonomous robots
- Integrating motors, sensors, microcontrollers, and wireless communication

AI @ GT Applied Research Project Team | Atlanta, GA

August 2025 - Present

AI/ML Engineer & Researcher

- Researching lifelong learning architectures for autonomous household robots
- Running simulations with Vision-Language-Action models in SAPIEN
- Designing uncertainty-aware arbitration between neural policies and LLM reasoning